

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

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Jaypee Institute of Information Technology, Noida
End Term Examination, ODD 2024
B. Tech. III Semester

Course Title: Data Structures
Course Code: 15B11CI311

Maximum Time: 2 hr.
Maximum Marks: 35

C210.1	Explaining abstract data types, memory allocation schemes and need of linear and non-linear data structures
C210.2	Applying and implementing various linear data structures like array, linked list, stack and queue in different problems and applications
C210.3	Analysing the performance of various sorting and searching techniques
C210.4	Demonstrating and implementing various operations like search, traverse, insertion, deletion etc. on different non-linear data structures
C210.5	Applying appropriate data structure to design an efficient solution for given and identified problem

Note: The question paper consists of 7 questions.

Q1. The T&P cell maintains (insert, delete, update) the record of offered packages to the students which are unique. It is desired to retrieve following information from the stored data.

1. Lowest Package
2. Highest Package
3. Median Package

Propose an efficient and customized non-linear data structure to store the packages of students so that above queries can efficiently be performed. Also write the algorithm(s) to perform above queries.

[C210.5 (Create Level), 8 Marks]

Q2. An integer array is initialized as $A = [10, 9, 2, 19, 23, 32, 4, 7, 46, 18]$. Write the updated values of the array after the function *fun* (...) is executed; low & high are leftmost & rightmost indexes of A.

[C210.2 (Apply Level), 3 Marks]

```
int fun (vector <int>&A, int low, int high)
{
    int p= A[high]
    int i= low-1;
    for (int j=low; j<high; ++j)
    {
        if (A[j] <=p)
        {
            ++i;
            swap (A [i], A[j]);
        }
    }
    swap (A [i+1], A[high])
    return i+1;
}
```

Q3. A binomial heap H3 is constructed through the union of two binomial heaps H1 and H2. H1 has 3 binomial trees of order 5, 6 & 8. H2 has 3 binomial trees of order 4, 5 & 9.

- a) Find the order of H3 assuming that H3 maintain binomial trees of unique degrees after union operation.
- b) Calculate the total number of nodes at level 3 of H3.

[C210.4 (Apply Level), 3 Marks]

Q4. Given a string $S = \text{"CALCUTTA"}$,

a) Create a suffix array Ar from string S for finding substring "CALC" in string S .

b) For searching the substring, show the intermediate steps and find the no. of times array Ar is accessed.

[C210.3 (Analyze Level), 5 Marks]

Q5. Given an expression containing different types of brackets $[, (, \{,],), \}$. Implement a function to check whether the expression is balanced or not utilizing both stack & queue.

[C210.2 (Apply Level), 6 Marks]

Q6. An integer array $A = [10, 9, 2, 19, 23, 32, 4, 7, 46, 18]$ is given. Draw the segment tree, showing indexes & values of each node for range sum query (RSQ) for any range $l : r$ on A . (l and r are left and right indexes of array A respectively)

[C210.4 (Apply Level), 4 Marks]

Q7. You are engineer tasked with optimizing a poorly designed communication network for 'n' integrated circuits (IC's) in a high-performance electronic system. The current system uses wires of various lengths to connect the IC's but it is prone to catastrophic failure. The current starts flowing from a specific IC. If current flows from any IC and loops back to the same IC, causes a short circuit.

Your task is to redesign the network by removing unnecessary connections to ensure no such looping paths exist while keeping all IC's connected either directly or indirectly. Additionally, the new design should minimize the total length of wires. Determine the minimum total length of wires required to transform the given network into safe and efficient one. Show all the intermediate steps and data structures used while optimizing the network given in Fig. 1.

[C210.5 (Create Level), 6 Marks]

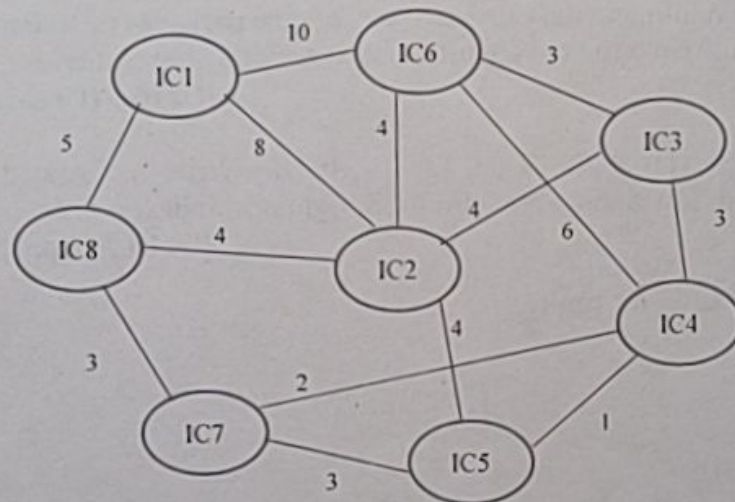


Fig. 1 Integrated Circuit